

Conclusively identified as **Saturnia pavonia**!

Classification result (scientific name, common name, confidence %)

1. Saturnia pavonia (Small Emperor Moth)	89.4%
2. Saturnia pavoniella (Ligurian Emperor)	3.3%

► Show thinking (5 steps)

Saturnia pavonia, commonly known as the Emperor Moth, is a species of moth belonging to the family Saturniidae. It is notable for its large size and striking appearance, with males typically having more vibrant colors than females. The wings are often adorned with eye-like patterns, which serve as a defense mechanism against predators.

Status in the UK

In the UK, **Saturnia pavonia** does not have an explicit conservation status, and it is not considered invasive. This means that while it is present in the region, it does not face significant threats that would require conservation measures, nor does it pose a risk to local ecosystems.

What to Do About It

If you encounter **Saturnia pavonia**, there are a few considerations:

- Observation:** Enjoy observing this beautiful moth in its natural habitat. They are often found in heathland, woodland edges, and gardens.
- Photography:** If you're interested in photography, this species provides excellent opportunities due to its striking appearance.
- Conservation Awareness:** While it currently has no conservation concerns, staying informed about local wildlife and their habitats is always beneficial.
- Avoid Disturbance:** If you find a caterpillar or a moth, avoid disturbing it, as they are part of the local ecosystem.

If you have specific concerns or need further information about interactions with this species, feel free to ask!

🚫 **This may be a regionally invasive species** — check local reporting/disposal rules before doing anything with it.

Sources:

- English Botany, 1863 (James Sowerby and James Edward Smith, 1790-1814. Replaces prior vague title -- Flora Britannica (1996, Richard Mabey) is NOT public domain, was a data error.) — <https://archive.org/details/englishbotanyorc12sowerich>
- The English Flora, Smith, James Edward, Sir, 1759-1828, [182 (James Edward Smith, 4 volumes, 1824-1828.) — <https://archive.org/details/englishflorasje01smit>

What is the scientific name of the small emperor moth and who first described it?

► Show thinking (2 steps)

The scientific name of the small emperor moth is **Saturnia pavonia**. It was first described by **Carl Linnaeus** in his 1758 work, the 10th edition of *Systema Naturae*.

Sources:

- British Butterflies, 1890 (W. S. Coleman, Routledge, 1860. Corrected title (was invented as A History of British Butterflies).) — <https://archive.org/details/historyofbritish03morr>
- British Entomology (John Curtis), Curtis, John, 1791-1862, 1824 (Illustrated early UK insect reference.) — <https://archive.org/details/britishentomolog01curt>
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella

What family does it belong to and where can it be found?

► Show thinking (2 steps)

The small emperor moth, **Saturnia pavonia**, belongs to the family **Saturniidae**. It can be found throughout the **Palearctic region** and is the only member of its family present in the British Isles, where it is commonly referred to simply as the emperor moth. The species typically inhabits a range of environments, including heathland and moorland.

Sources:

- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- British Entomology (John Curtis), Curtis, John, 1791-1862, 1824 (Illustrated early UK insect reference.) — <https://archive.org/details/britishentomolog01curt>
- Wikipedia, "Cydalima perspectalis" — https://en.wikipedia.org/wiki/Cydalima_perspectalis
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella

What are the main differences in appearance between males and females?

► Show thinking (2 steps)

The main differences in appearance between male and female **Saturnia pavonia** (emperor moth) are as follows:

- Size:** Females are generally larger than males, being about one-third bigger.
- Coloration:**
 - Males:** The forewings are typically a chestnut-brown with a distinctive white band and a black eyelet with a lilac pupil. The hindwings are pale chestnut.
 - Females:** The forewings have a more extensive black area at the apex and termen, and the hindwings feature a larger black spot before the apex. The underside of the forewings is light ochraceous yellow, while the hindwings are also lighter compared to the males.
- Markings:** The males have a more pronounced white band across the forewings, while the females have larger black discal spots on the forewings.

These differences help in identifying the sexes of the emperor moth in the field.

Sources:

- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- British Butterflies, 1890 (W. S. Coleman, Routledge, 1860. Corrected title (was invented as A History of British Butterflies).) — <https://archive.org/details/historyofbritish03morr>
- British Entomology (John Curtis), Curtis, John, 1791-1862, 1824 (Illustrated early UK insect reference.) — <https://archive.org/details/britishentomolog01curt>
- Wikipedia, "Pieris brassicae" — https://en.wikipedia.org/wiki/Pieris_brassicae

When do males fly and what do they look for?

► Show thinking (2 steps)

Males of **Saturnia pavonia** (emperor moth) typically fly during the day from mid-April to late June. They are known for their rapid flight as they search for females, which are generally more sluggish and tend to fly at night.

In terms of appearance, male emperor moths have a wingspan of about 60 mm (2.4 inches) and are characterized by their brown and white forewings, which are marked with red and orange fascia and a bold black and orange eyespot. The hindwings are orange with a similar eyespot, making them visually distinct during flight.

Sources:

- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- British Butterflies, 1890 (W. S. Coleman, Routledge, 1860. Corrected title (was invented as A History of British Butterflies).) — <https://archive.org/details/historyofbritish03morr>
- British Entomology (John Curtis), Curtis, John, 1791-1862, 1824 (Illustrated early UK insect reference.) — <https://archive.org/details/britishentomolog01curt>
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella

What habitats does this moth prefer?

► Show thinking (2 steps)

Saturnia pavonia (emperor moth) prefers a variety of habitats, but it is most commonly associated with heathland and moorland. These environments provide the necessary conditions for both the adult moths and their larvae, which feed on a wide range of plants, with heather being the most common food source. The species can also be found in gardens and woodland edges, where suitable host plants are available.

Sources:

- Wikipedia, "Aconitum" — <https://en.wikipedia.org/wiki/Aconitum>
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella

What does the caterpillar look like and what does it eat?

► Show thinking (2 steps)

The caterpillar of **Saturnia pavonia** (emperor moth) exhibits a striking appearance that can vary. Initially, it is described as being black and orange, but as it matures, it can turn green with black rings and may have yellow and red spots along its body.

In terms of diet, the caterpillars are known to feed on a wide variety of plants. The most common food source is heather (Calluna), but they have also been recorded feeding on other plants such as birch (Betula), hawthorn (Crataegus), and various species of willow (Salix), among others. This diverse diet supports their growth and development before they pupate.

Sources:

- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- British Butterflies, 1890 (W. S. Coleman, Routledge, 1860. Corrected title (was invented as A History of British Butterflies).) — <https://archive.org/details/historyofbritish03morr>
- Wikipedia, "Cinnabar moth" — https://en.wikipedia.org/wiki/Cinnabar_moth
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella

What is a common incorrect genus name sometimes used for this species?

► Show thinking (2 steps)

A common incorrect genus name sometimes used for **Saturnia pavonia** is **Saturnia** itself, as it is often mistakenly referred to in various contexts. However, the correct genus name is **Saturnia**, and this confusion can arise in literature or discussions about the species.

Sources:

- A Handbook of the British Flora, 1887 (George Benthams, 1858.) — <https://archive.org/details/handbookofbrit00bent>
- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- English Botany, 1863 (James Sowerby and James Edward Smith, 1790-1814. Replaces prior vague title -- Flora Britannica (1996, Richard Mabey) is NOT public domain, was a data error.) — <https://archive.org/details/englishbotanyorc12sowerich>
- The English Flora, Smith, James Edward, Sir, 1759-1828, [182 (James Edward Smith, 4 volumes, 1824-1828.) — <https://archive.org/details/englishflorasje01smit>

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